Higher National Diploma in Civil Engineering
Fourth Year, First Semester Examination – 2018
CE 3221 – Highway and Railway Engineering

Instructions for Candidates:	No. of questions	: 06
Answer any **Five (05)** questions.	No. of pages	: 03
All question carry equal marks	Time	: Three (03) hours

1.

a) What do you mean by Geometrical elements of a highway? (05 marks)

b) Draw simple diagrams of simple, compound, and reverse curves. (05 marks)

c) How to find the deflection angle when radius and chord length are given in a curved section. (05 marks)

d) What do you mean by intersection point of a simple cure and write a relationship in between tangent length, radius of the curve and deflection angle made by two tangents. (05 marks)

2.

a) A one kilometer long section of A1 road (2lane) with the curve of 100 m radius is to be improved level of series (design speed of 70km/h). Road cross section consist of 3.5m lane, 0.6m shoulders, and 0.6m drain (Assume side friction of 0.2, longitudinal friction coefficient is 0.29 and gradient is zero)

I. Determine the super elevation rate you recommend the design speed of 70km/h. (04 marks)

II. Determine the minimum stopping sight distance at the design speed. (04 marks)

b) Presuming you are the alignment engineer for a road design project & you have been asked to design a horizontal curve to left by using the minimum radius to suit a design speed of 70 km/h & for a super elevation and side friction are respectively 3% and 0.17. Assume point of intersection of two tangents has a station 2+020 & the deflection angle is 21°31'53".

I. Calculate the minimum radius for the curve. (04 marks)

II. Calculate the length of the curve. (04 marks)

III. Calculate chainages at Tangent to Curve (TC) & Curve to Tangent (CT). (04 marks)

3.

a. Defines the following terms with respect to roads & highways.
   I. Camber
   II. Gradient
   III. Right of way
   IV. Sight distance (08 marks)

b. Explain the term "Super elevation" and it is importance in high Way designing. (04 marks)

c. What do you mean by transition curves in highways and it is Importance of high way designing. (04 marks)

d. Discuss the use of mass haul curves in road project. (04 marks)

4.

a. When selecting suitable type of gravel for road work which type of tests need to be done initially & explain the reasons. (04 marks)

b. When selecting gravel sample for testing from a gravel quarry which Method need to be applied. (04 marks)

c. Explain the method need to be followed to choose the sample which is going to test in the laboratory for collected gravel from the quarry. (04 marks)

d. Explain the reason why bitumen cannot be applied as binding material of gravel. (04 marks)

e. Explain the method of selecting of suitable moisture content in the field for compaction of loose gravel base. (04 marks)

5.

a. Draw a complete cross section of a modern type of highway with identifying their existing components. (04 marks)

b. What do you mean by type A, B, C, D, & E Roads in Srilanka? (04 marks)

c. Explain the following terms by considering road safety.
   I. Stopping distance
   II. perception time
   III. Brake reaction time
   IV. Brake efficiency
   V. lag distance (10 marks)

d. Explain the importance of Appling tack coat for surface treatments of roads. (02 marks)

6.

a. What do you mean by interrupted traffic flow? (02 marks)

b. Defined the following in the traffic streams
   I. Car following
   II. lane changing
   III. overtaking

IV. merging
V. Diverging
VI. Queuing

(06 marks)

c. What do you mean by traffic volume? (03 marks)

d. Define following traffic terms
   I. Hourly traffic volume (HTV)
   II. Average daily traffic (ADT)
   III. Average Annual daily traffic (AADT)
   IV. Peak hour traffic volume (PHV)

(04 marks)

e. Define the following terms
   I. Spot speed
   II. Running speed
   III. Journey speed
   IV. Space mean speed
   V. Time mean speed

(05 marks)